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SIMULATION EXPERIMENT – 5

Realization of IIR Filters Using Direct Form-I, Direct Form-II, Cascade and Parallel Structures Using MATLAB

MATLAB CODE:

```
% Experiment: Realization of IIR Filters Using
% Direct Form-I, Direct Form-II, Cascade and Parallel Structures

clc;
clear;
close all;

%% Step 1: Design an IIR Butterworth Low Pass Filter
N = 4;              % Filter Order
Fc = 0.4;           % Normalized Cutoff Frequency
[b,a] = butter(N,Fc);
disp('Numerator Coefficients');
disp(b);
disp('Denominator Coefficients');
disp(a);

%% Step 2: Input Signal
n = 0:50;
x = [1 zeros(1,50)];      % Unit Impulse Signal
%% Step 3: Direct Form-I / Direct Form-II Realization
% MATLAB filter() realizes the transfer function
y_df = filter(b,a,x);

%% Step 4: Cascade Realization (Second Order Sections)
```

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```
[sos,g] = tf2sos(b,a);
disp('Second Order Sections');
disp(sos);
y_cascade = sosfilt(sos,x) * g;

%% Step 5: Parallel Realization
[r,p,k] = residuez(b,a);
disp('Residues');
disp(r);
disp('Poles');
disp(p);
disp('Direct Term');
disp(k);
[bp,ap] = residuez(r,p,k);
y_parallel = filter(bp,ap,x);

%% Step 6: Plot Responses
figure;
subplot(2,2,1)
stem(n,y_df,'filled');
title('Direct Form I / II');
xlabel('Samples');
ylabel('Amplitude');
grid on;

subplot(2,2,2)
stem(n,y_cascade,'filled');
title('Cascade Structure');
```

```
xlabel('Samples');
ylabel('Amplitude');
grid on;

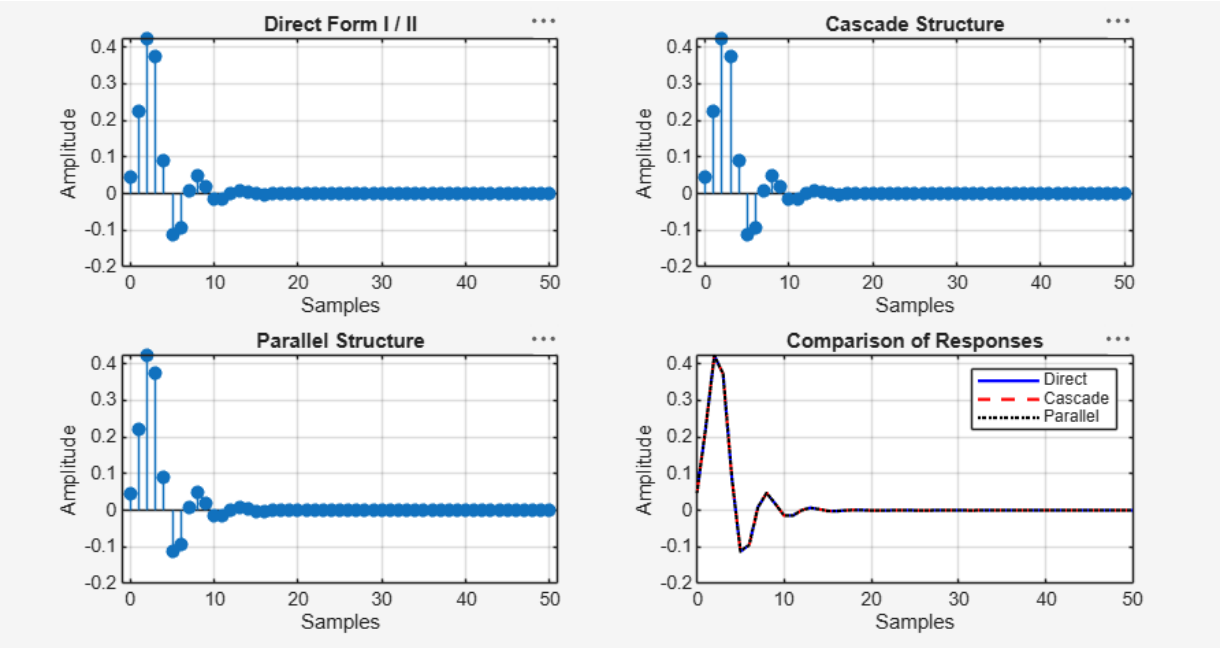
subplot(2,2,3)
stem(n,y_parallel,'filled');
title('Parallel Structure');
xlabel('Samples');
ylabel('Amplitude');
grid on;

subplot(2,2,4)
plot(n,y_df,'b','LineWidth',1.5);
hold on;
plot(n,y_cascade,'r--','LineWidth',1.5);
plot(n,y_parallel,'k:','LineWidth',1.5);
legend('Direct','Cascade','Parallel');
title('Comparison of Responses');
xlabel('Samples');
ylabel('Amplitude');
grid on;

%% Step 7: Verification
disp('Maximum Difference (Direct vs Cascade)');
disp(max(abs(y_df-y_cascade)));
disp('Maximum Difference (Direct vs Parallel)');
disp(max(abs(y_df-y_parallel)));
```

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OUTPUT




```
Numerator Coefficients
    0.0466    0.1863    0.2795    0.1863    0.0466

Denominator Coefficients
    1.0000   -0.7821    0.6800   -0.1827    0.0301

Second Order Sections
    1.0000    2.0000    1.0000    1.0000   -0.3290    0.0646
    1.0000    2.0000    1.0000    1.0000   -0.4531    0.4663

Residues
  -0.4039 + 0.3123i
  -0.4039 - 0.3123i
  -0.3461 - 2.3795i
  -0.3461 + 2.3795i

Poles
   0.2266 + 0.6442i
   0.2266 - 0.6442i
   0.1645 + 0.1937i
   0.1645 - 0.1937i

Direct Term
    1.5466

Maximum Difference (Direct vs Cascade)
   1.2212e-15

Maximum Difference (Direct vs Parallel)
   3.8858e-16
```

RESULT

The given IIR transfer function was successfully realized using **Direct Form-I, Direct Form-II, Cascade, and Parallel structures** in MATLAB. The coefficients, impulse responses, and output responses obtained from all realizations were found to be nearly identical, confirming that each structure implements the same transfer function with only negligible numerical differences. Thus, the objectives of realizing, comparing, and verifying the different IIR filter structures were successfully achieved, and the expected outcomes were attained.